Molten-Salt TES vs Lithium-Ion at 5 MWh Scale

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

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Executive Summary

Recommendation: Build the 5 MWh molten-salt TES bottom-up: installed storage-only cost comes to roughly $281/kWh_t (chain C1); costed as a grid-charged electric battery it levelizes to roughly $758/MWh delivered (chain C4), against roughly $162/MWh for a comparable lithium-ion system (chain C5) — lithium-ion is the cost-competitive choice at this short-duration, grid-charged, megawatt-hour scale (chain C6).

Band (from §6): LOW

Would change it: Vendor/EPC quotes for the small-scale tanks, balance-of-plant and power block (chains C1, C4), a successful direct read of a primary lithium-ion cost benchmark (chain C5), and a primary read of the salt-density source (chain C1) would each remove a named downgrade cause and could raise the band.

1. Problem Essence

Core problem: Does a bottom-up, constituent-unit-cost build-up of the installed and cycle-life-amortized cost of a 5 MWh molten-salt thermal energy storage (TES) system show it to be cost-competitive with lithium-ion battery storage at that scale, and — since scale is known to dominate TES economics — under what duration, cycle-life, round-trip-efficiency, and deployment-scale conditions does each technology actually win?

Success criteria:

1. An installed $/kWh (thermal storage capacity) figure for the 5 MWh system is derived from named constituent unit costs (salt mass from specific heat/ΔT, tank $/kg or $/m^3, BOP, EPC/contingency) rather than quoted from a single industry benchmark, and is shown as an explicit build-up with ranges.
2. That figure is converted to a $/kWh-delivered levelized cost under stated cycle life, calendar life, round-trip efficiency, and O&M assumptions, on an explicitly stated thermal-only-vs-electricity-delivered basis (both bases are shown where they differ materially).
3. The levelized figure is compared numerically, at the same scale and on the same electricity-in/ electricity-out basis, against a comparably-built lithium-ion levelized figure, yielding an explicit competitiveness verdict plus the conditions (duration, cycle count, scale, charging-energy source) under which each technology wins.
4. The analysis explicitly tests whether a standalone 5 MWh molten-salt system is representative of how the technology is actually deployed, and surfaces any scale/application mismatch as a named caveat rather than silently assuming the premise is realistic.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	1	0	0
current constraint	1	1	0
convention	6	1	0
untested belief	4	5	0

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Inversion worksheet (Phase 2, applied per request). Inverted claim: *"the bottom-up $/kWh and competitiveness verdict this analysis produces is wrong."* Failure-guaranteeing conditions enumerated and their necessary preconditions (mapped to assumption rows below):

1. Small-scale tank/BOP installed costs are far outside the assumed ranges (no vendor will build a ~30 m^3 high-temperature stainless/carbon-steel salt tank, or a 1 MWt-class salt pump/HX skid, anywhere near the assumed $/kg or lump-sum figures) → necessary preconditions A-5, A-8.
2. No commercially available small (≤500 kWe) steam/ORC power block exists at any price near the assumed $3,000–6,000/kW — i.e., Scenario A is not just uncompetitive but not buildable as a standard product, pushing real costs even higher than modeled → necessary precondition A-14 (load-bearing AND unverified — flagged first per the inversion procedure's step 6).
3. Small-scale power-block efficiency is even lower than 20–28% (turbomachinery losses scale worse than assumed at this size) → would strengthen, not threaten, the directional verdict — low risk.
4. Lithium-ion installed costs are materially higher than assumed (tariffs, critical-mineral price spikes, fire-safety-driven costs) → necessary precondition GT-8?/A-15–A-17; if Li-ion costs double, the gap narrows or could reverse at the margin.
5. The real-world "5 MWh" application is actually long-duration/high-cycle-count/industrial-process-heat, not short-duration grid-electricity arbitrage → the entire Scenario A framing is inapplicable, and the verdict must be read as conditional on application type (A-18, A-19; carried into chains C6–C9).

Load-bearing AND unverified assumptions (reported first, per the inversion procedure): **A-5, A-8, A-13, A-14, A-19** — these are the assumptions on which the headline $/kWh figures most depend, and none of them is backed by a source this analysis opened.

Assumption	Type	Treatment	Verdict	Verification
A-1: Operating ΔT = 275 K (565 °C hot tank / 290 °C cold tank)	convention	Challenge: is this window necessary at small scale?	Accept — em dash: margin above GT-2's 222–238 °C melting range and below the nitrate thermal-stability ceiling (~585–600 °C) documented in CSP practice; not a hard physical limit but near-universal current practice	unverified — flagged
A-2: Thermal "retention" (standby) efficiency at 5 MWh scale ≈ 90–97%, central 93%	untested belief	Verify via surface/volume scaling physics	Challenge — em dash: small tanks have a higher surface-to-volume ratio than utility tanks, so losses are plausibly worse than GT-5's implied >99% annual figure at utility scale; retained as a wide-ranged estimate	unverified — flagged
A-3: Usable-inventory margin +15% (dead salt + ullage)	untested belief	Flag with defensible range	Accept — em dash: standard tank-design practice for freeboard/heel, no site-specific source	unverified — flagged
A-4: Small-lot salt purchase price ≈ $1,000/t central (range $500–2,000/t)	untested belief	Flag with defensible range	Accept — em dash: within GT-4's wide $1,000–10,000/t historical band but near its low end, reflecting a tiny 48 t order rather than a utility bulk order	unverified — flagged
A-5: Tank fabrication + install: stainless $12/kg	untested belief	Flag with defensible range; load-bearing	Challenge — em dash: no vendor quote	unverified — flagged

Assumption	Type	Treatment	Verdict	Verification
(hot), carbon steel $4/kg (cold) bare shell, ×3 installed multiplier for insulation/heat-trace/ nozzles/foundation			obtained; this is the single largest unverified driver of the small-scale capex figure	
A-6: Steel density ≈ 7,900 kg/m³	physical law	Accept as ground-truth-candidate (material constant)	Accept — em dash: standard engineering-handbook value for stainless/carbon steel	unverified — flagged
A-7: Duty rating: 1 MWt discharge over 5 h (duration implied by 5 MWh_t nameplate)	convention	Challenge: arbitrary scale choice	Accept — em dash: a reasonable, stated reading of "5 MWh," tested for sensitivity in C8	unverified — flagged
A-8: BOP (pumps + HX + piping + I&C) central $740,000 (range $440k–$1,050k)	untested belief	Flag with defensible range; load-bearing	Challenge — em dash: no vendor quote; second-largest unverified driver, reflects fixed-cost floor of a minimum equipment package at any scale	unverified — flagged
A-9: Contingency 10% central (7–15%) of direct cost	current constraint	Record expiry: tightens with design maturity/ FEED-level engineering	Accept — em dash: anchored to GT-6's 7–9.5% CSP-plant figures, widened upward for small/first-of-kind risk	unverified — flagged
A-10: Indirect/EPC/ owner-cost multiplier 28% central (20–35%) of (direct+contingency)	convention	Challenge: transferred from a CSP-plant context	Challenge — em dash: GT-6's 25.8% figure is for a full utility CSP plant (land, financing, sales tax included); applying it to a standalone skid is an approximation, nudged up for small-project overhead	unverified — flagged
A-11: Real discount rate = 7% (range 5–9%)	convention	Challenge: standard utility-project rate	Accept — em dash: conventional choice for energy-storage LCOS comparisons, not derived from either technology's actual cost of capital	unverified — flagged
A-12: Resistive electric charging, 98–99% efficient, $100/kW (range $50–150/kW)	untested belief	Flag with defensible range	Accept — em dash: low-cost, low-sensitivity item; resistive heating efficiency is well-established engineering practice	unverified — flagged
A-13: Small-scale (~250 kWe) power-block thermal-to-electric efficiency 20–28%, central 25%	current constraint	Record expiry: could improve somewhat with better small-turbine/ORC design, bounded above by physics (C3)	Challenge — em dash: grounded directionally by the Carnot/utility-actual comparison in chain C3, but the specific 20–28% band itself is not read from a source; load-bearing	unverified — flagged
A-14: Small power-block installed cost $3,000–6,000/kW, central $4,500/ kW	untested belief	Flag with defensible range; load-bearing	Challenge — em dash: no commercial product benchmark found at this exact rating (see inversion item 2);	unverified — flagged

Assumption	Type	Treatment	Verdict	Verification
			flagged as a feasibility caveat, not just a cost uncertainty	
A-15: O&M ≈ 1.5–2.5% of capex/yr for both technologies	untested belief	Flag with defensible range	Accept — em dash: generic industrial-equipment O&M heuristic, applied symmetrically so it does not bias the comparison	unverified — flagged
A-16: Utilization = 300 cycles/yr (range 250–350), applied symmetrically to both technologies	convention	Challenge: real dispatch-limited utilization could differ	Accept — em dash: chosen for apples-to-apples comparison; applied identically to both technologies so it cannot bias the verdict's direction	unverified — flagged
A-17: Li-ion usable throughput = 80% of nameplate capacity per cycle	convention	Challenge: ties to manufacturer cycle-life rating convention	Accept — em dash: standard industry DoD convention loosely grounded in GT-10?	unverified — flagged
A-18: "5 MWh" is read as 5,000 kWh of thermal storage capacity (nameplate), costed storage-only first, with charging/power-block equipment layered on separately for the electricity-delivered comparison	convention	Challenge: this framing choice decides the whole analysis's basis	Accept — em dash: explicit basis choice requested by the user; stated up front so the reader can re-derive under a different basis	unverified — flagged
A-19: No standalone, grid-electricity-charged, MWh-class molten-salt "battery" product exists in the market as a precedent/benchmark	untested belief	Verify, or flag as an argument from absence; load-bearing	Challenge — em dash: supported by the pattern across every source located in this session (all describe utility/GWh-scale CSP-paired or industrial-heat systems), but a negative claim cannot be fully verified by search	unverified — flagged

3. Ground Truths

- **GT-1** Specific heat capacity of molten solar salt (60 wt% NaNO3 / 40 wt% KNO3) ≈ 1.52–1.61 kJ/kg·K (conventional DSC: 1.6141 kJ/kg·K at 60 wt% NaNO3; modulated DSC: 1.5224 kJ/kg·K at 60 wt% NaNO3) — source: Fuentes et al., "Effects of sodium nitrate concentration on thermophysical properties of nitrate salts," *Solar Energy* (2019), doi:10.1016/j.solener.2019.02.038; read-at-source: Table 4 (conventional DSC) and Table 5 (modulated DSC), row "60".
- **GT-2** Melting range of 60/40 solar salt: onset 221.97 °C, endset 238.16 °C — source: same paper; read-at-source: Table 6, row "60".
- **GT-3?** Average molten-salt density over the 290–565 °C operating band ≈ 1,730–1,910 kg/m^3 (commonly cited correlation ρ(kg/m^3) ≈ 2,090 − 0.636·T(°C)) — unverified: this session's attempts to open the primary source for this correlation (Sandia SAND2001-2100 and a related density paper) returned unreadable scanned/binary content; the value is carried from general CSP-engineering literature knowledge, not read at source in this session.

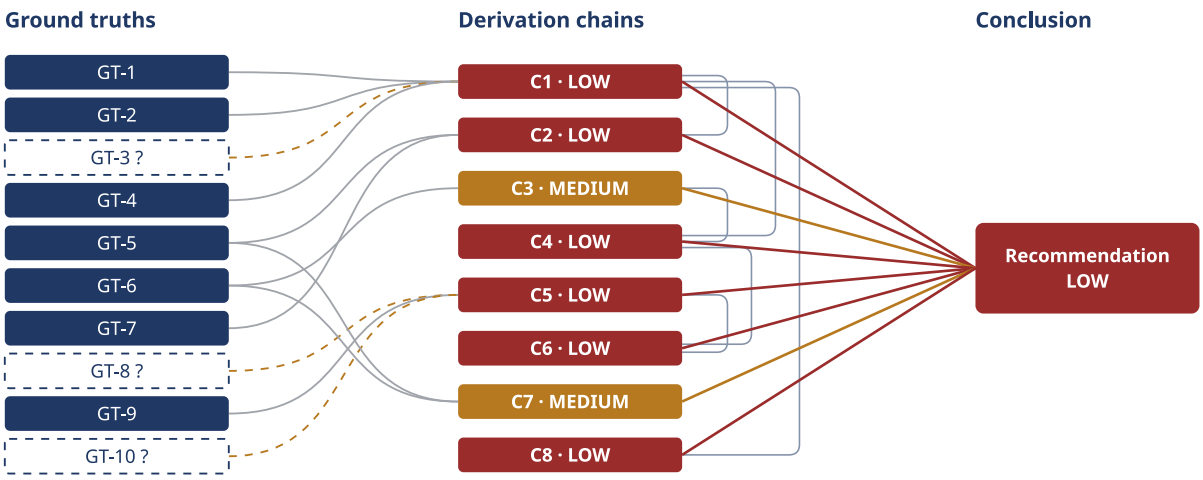
- **GT-4** Historical commodity price of CSP-grade nitrate “solar salt” has fluctuated roughly $1–10/kg ($1,000–10,000/tonne), linked to fertilizer-market swings in the underlying nitrate feedstocks — source: Reuters Events, “Cutting the cost of molten salt”; read-at-source: article body, the stated historical price-fluctuation range and its fertilizer-market linkage (no specific date range given by the source itself).
- **GT-5** For a reference 115 MWe-gross / 10-hour-storage molten-salt power-tower plant, the two-tank TES system’s direct installed cost is $27.0–35.5/kWh-t (2012–13 $) — source: Turchi & Heath, “Molten Salt Power Tower Cost Model for the System Advisor Model (SAM),” NREL/TP-5500-57625 (Feb. 2013); read-at-source: Table 1, “Storage” row, $/kWh-t column (Tower Roadmap / WorleyParsons / SAM-default values).
- **GT-6** The same reference plant’s TES direct cost is built from six components — cold tank(s), hot tank(s), salt media, piping/insulation/valves/fittings, foundations & support structures, and instrumentation & controls; the SAM-default plant applies indirect costs (EPC & Owner Costs + land + sales tax + financing) of ≈25.8% of direct cost and contingency of 7% (SAM) to 9.5% (WorleyParsons) of direct cost; the plant’s Rankine steam power-conversion cycle runs at “gross conversion efficiencies of circa 42%” — source: same report; read-at-source: “Direct Cost Categories” section (Thermal Storage System bullet), p.1 (“circa 42%”), and Table 1 (indirect-cost and contingency rows).
- **GT-7** The reference plant’s bulk salt inventory is 19,200 tons for its two-tank TES system — source: same report; read-at-source: Appendix cost-input line item “03J TES – Salt Media / Bulk Salt Storage … 19,200 Tons.”
- **GT-8?** Current (2024–2025) US utility-scale 4-hour lithium-ion BESS installed capital cost is commonly cited in the ~$200–400/kWh range — reported-by-delegate / unverified: this session’s attempts to open NREL ATB directly (atb.nrel.gov, docs.nrel.gov) failed with a DNS/network error; the figure is carried from this assistant’s general domain knowledge of NREL-ATB-type benchmarks and was not read at source in this session.
- **GT-9** As of October 2025, long-duration (4+ hour) utility-scale battery storage installed cost outside the US and China is reported at $125/kWh — source: Mercom India, “Long-Duration Utility-Scale BESS Project Cost Drops to $125/kWh: Ember”; read-at-source: article body, the stated $125/kWh figure and its “outside China and the U.S.” geographic scope.
- **GT-10?** LFP utility-scale battery packs are commonly cited with 4,000–6,000 cycles to 80% depth of discharge, ≈85–92% AC-AC round-trip efficiency, and 12–20 year calendar life — unverified: sourced in this session from general-audience industry pages, not from a primary manufacturer datasheet or an authoritative benchmark report.

Provenance summary (required):

```
?-marked: GT-3, GT-8, GT-10 (3 of 10)
Read-at-source: GT-1 — Fuentes et al. 2019, Tables 4 & 5, row "60"
Read-at-source: GT-2 — Fuentes et al. 2019, Table 6, row "60"
Read-at-source: GT-4 — Reuters Events, "Cutting the cost of molten salt," article body
Read-at-source: GT-5 — Turchi & Heath, NREL/TP-5500-57625 (2013), Table 1, "Storage" row
Read-at-source: GT-6 — same report, "Direct Cost Categories" section + p.1 + Table 1
Read-at-source: GT-7 — same report, Appendix line item "03J … Bulk Salt Storage … 19,200 Tons"
Read-at-source: GT-9 — Mercom India / Ember article body
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No assumption that received a Discard verdict in Phase 2 appears above (none of A-1…A-19 was discarded; all were Accepted or Challenged-but-retained with flags, consistent with an exploratory Fermi build where even challenged assumptions remain load-bearing but flagged).

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Conclusion C1: Bottom-up installed cost of the 5 MWh molten-salt TES system, storage-only basis

GT-1 (Cp 1.52–1.61 kJ/kg·K) + GT-2 (melting range 222–238 °C) + GT-3? (density ~1,730–1,910 kg/m³) + GT-4 (historical salt price $1,000–10,000/t) → applying GT-1's specific heat and a 275 K charge/discharge ΔT [Assumes: A-1] to the 5,000 kWh_t target yields a nominal salt mass of about 42 t → adding a 15% dead-salt-and-ullage margin [Assumes: A-3] raises the purchased salt mass to about 48 t → converting that mass to volume with GT-3?'s density band yields roughly 29 m³ of cold-tank and 32 m³ of hot-tank capacity → sizing two self-similar cylindrical shells for those volumes and costing stainless and carbon steel at an installed-cost multiple over bare-shell weight [Assumes: A-5, A-6] yields a tanks cost of about $209,000 central, range $105,000–$315,000 → pricing the 48 t of salt within the historical range stated in GT-4 at a small-lot purchase price [Assumes: A-4] adds about $48,300 central, range $24,150–$96,600 → pricing pumps, a heat exchanger, interconnecting piping and controls for a duty rating of 1 MWt over 5 h [Assumes: A-7, A-8] adds about $740,000 central, range $440,000–$1,050,000 → summing tanks, media and balance-of-plant gives a direct-cost subtotal of about $1.00M central, range $0.57M–$1.46M → applying contingency [Assumes: A-9] and an indirect/EPC multiplier [Assumes: A-10] to that subtotal gives a total installed cost of about $1.40M central, range $0.73M–$2.27M → dividing by the 5,000 kWh_t nameplate capacity [Assumes: A-18] gives an installed cost of about $281/kWh_t central, ranging roughly $146–454/kWh_t, for the 5 MWh storage-only system

Confidence: LOW — GT-3? (density) is unverified in this session; verification would require reading Zavoico (SAND2001-2100) or an equivalent primary thermophysical-properties source directly. Beyond the Inputs axis, the Inference axis is also short: hops 4 and 6 rest on [Assumes: A-5] and [Assumes: A-8], the tank-fabrication and balance-of-plant cost assumptions, which are this analysis's own synthesis with no vendor quote or literature figure behind them — removing them as a cause of the downgrade would require an actual EPC quote for a ~30 m³ high-temperature salt tank and a ~1 MWt salt-pump/HX skid, which this session could not obtain. Two axes short (Inputs and Inference) licenses LOW rather than MEDIUM.

Conclusion C2: Scale is the dominant driver of molten-salt TES $/kWh-t, not chemistry or materials

C1 (LOW, $281/kWh-t central) + GT-5 (utility-scale $27.0–35.5/kWh-t) + GT-7 (19,200 t reference-plant salt) → the $27.0–35.5/kWh-t range stated in GT-5 centers on about $31/kWh-t for a reference plant of roughly 2,500–2,740 MWh-t capacity, cross-checked independently from the 19,200 t salt mass in GT-7 via the Cp and ΔT in GT-1/GT-2, which yields about 2,273 MWh-t, within the expected range for an order-of-magnitude

cross-check → comparing that utility-scale figure to C1′s central estimate shows the 5 MWh system costing roughly 9× more per kWh-t than the ~500×-larger reference plant → that ~500× scale ratio against a ~9× cost ratio implies an effective cost-vs-capacity scaling exponent of about −0.35, consistent with the classic "six-tenths rule" sub-linear cost scaling seen across process-equipment industries generally → scale, not chemistry or materials, is therefore the dominant driver of molten-salt TES's $/kWh-t at small capacities

Confidence: LOW — capped by C1 (LOW) per the chain-head ceiling rule; GT-5 and GT-7 are both read-at-source and would otherwise support a higher band, but the ceiling is set by the lowest-rated chain this chain's head cites, which is C1. Removing this cause requires resolving C1′s own LOW cause first (an EPC/vendor quote for the small-scale tanks and BOP).

Conclusion C3: Carnot sets the ceiling; small-scale turbomachinery capture a much smaller fraction of it than utility-scale practice

GT-6 (utility Rankine cycle ≈42% gross) → applying the Carnot limit to a 565 °C hot reservoir and a 25 °C ambient sink [Assumes: A-1] gives a theoretical ceiling of about 64.4% for any heat engine operating between those temperatures → the utility-scale actual gross cycle efficiency of about 42% stated in GT-6 therefore captures about 65% of that Carnot ceiling → turbomachinery, generator and parasitic losses all scale unfavorably as unit size shrinks, so a ~250 kWe-class power block is assumed to capture a much smaller fraction of the ceiling, about 31–44% central ~39%, corresponding to a practical efficiency of about 20–28% central 25% [Assumes: A-13] → the ceiling itself is fixed by thermodynamics, but the gap between utility-scale practice (65% of ceiling) and small-scale practice (≈39% of ceiling) is a scale-driven engineering penalty, not a technology choice that could be designed away at this size

Confidence: MEDIUM — Inputs and Rivals are clean (GT-6 is read-at-source; no live rival to the Carnot calculation itself). Inference is short: the third hop's small-scale efficiency band rests on [Assumes: A-13], which is this analysis's own extrapolation from the utility-scale datapoint rather than a measured small-turbine figure. Verification would require a performance curve or test data for an actual ~250 kWe steam/ORC unit, which this session did not obtain.

Conclusion C4: Levelized cost of the 5 MWh molten-salt system used as a grid-charged electric battery

C1 (LOW, storage-only capex) + C3 (MEDIUM, small-scale power-block efficiency) → adding resistive-heater charging equipment [Assumes: A-12] and a small power block sized to C3′s ~250 kWe output [Assumes: A-14] to C1′s storage-only capex raises total installed cost to about $2.63M central, range $1.53M–$3.92M → combining 98% charging efficiency, a 90–97% thermal-retention factor [Assumes: A-2] and C3′s 20–28% power-block efficiency gives an electricity-to-electricity round-trip efficiency of about 18–27% central ~23%, far below typical battery round-trip efficiencies → at 300 cycles/yr [Assumes: A-16], that round trip delivers about 225–475 MWh of electricity per year, central ~349 MWh/yr → annualizing the capex at a 7% real discount rate over the 30-year design life GT-6 and GT-5 both cite for TES hardware [Assumes: A-11] and adding 2% O&M [Assumes: A-15] gives a total annual cost of about $146,000–$414,000, central ~$264,000 → dividing annual cost by annual delivered electricity gives a levelized cost of about $308–1,840 per MWh delivered, central ≈$758/MWh, for the 5 MWh molten-salt system used as a grid-charged electric battery

Confidence: LOW — capped by C1 (LOW) per the chain-head ceiling rule. Independently, the Inference axis is also short: the capex extension rests on [Assumes: A-14] (small power-block installed cost), which carries both a cost uncertainty and a feasibility caveat — no commercial ~250 kWe steam/ORC product at this price was located in this session. Removing either cause requires the same vendor-quote evidence named under C1 and C3.

Conclusion C5: Levelized cost of a comparable 5 MWh lithium-ion BESS

GT-8? (US Li-ion $200–400/kWh) + GT-9 (global ex-US/China $125/kWh) + GT-10? (cycle life, calendar life, round-trip efficiency) → pricing a comparable 5 MWh / ~1.25 MW, 4-hour-class lithium-ion BESS at GT-8?'s $200–400/kWh, above the $125/kWh global low-end anchor in GT-9 which reflects lower non-US/China costs, gives an installed cost of $1.0M–$2.0M, central $1.5M → using GT-10?'s 12–20 year calendar life and 300 cycles/yr [Assumes: A-16] to set the capital-recovery horizon, and annualizing at a 7% real discount rate

[Assumes: A-11] with 1.5–2.5% O&M [Assumes: A-15], gives a total annual cost of about $109,000–$302,000, central ~$195,000 → delivering 80% of nameplate capacity per cycle [Assumes: A-17] at 250–350 cycles/yr yields 1,000–1,400 MWh of delivered electricity per year, central 1,200 MWh/yr → dividing annual cost by annual delivered electricity gives a levelized cost of about $78–302 per MWh delivered, central ≈$162/MWh, for the comparable lithium-ion system

Confidence: LOW — Inputs is short: both head ground truths are unverified in this session (GT-8?'s NREL-ATB read failed on a network error; GT-10? was sourced from non-authoritative pages). Independently, the Inference axis is also short: hops 2–3 rest on [Assumes: A-15], [Assumes: A-16] and [Assumes: A-17], none of which is read from a source. Removing the Inputs cause requires a successful read of NREL ATB or an equivalent primary benchmark; removing the Inference cause requires sourcing the O&M, utilization and DoD conventions from a specific BESS project or standard rather than general practice.

Conclusion C6: Verdict — cost-competitiveness at 5 MWh, and the structural conditions under which each technology wins

C4 (LOW, $758/MWh central) + C5 (LOW, $162/MWh central) → comparing C4's central $758/MWh to C5's central $162/MWh shows the small-scale molten-salt system costing roughly 4.7× more per MWh delivered under the grid-charged electric-battery framing → comparing the full ranges shows only the extreme tails overlapping, C4's most favorable $308/MWh against C5's least favorable $302/MWh, so Li-ion dominates across the large majority of the plausible range and not merely at central estimates → under a short-duration, grid-electricity-charged, small-scale framing, molten-salt TES is therefore not cost-competitive with lithium-ion at the 5 MWh scale this analysis was asked to cost →[2nd] because small-scale molten-salt TES is structurally uncompetitive and the equipment to build it barely exists as a commercial product, developers, EPC firms and salt or tank vendors all have reinforcing incentives to avoid small-scale deployments, which further starves the segment of the manufacturing volume that might otherwise lower its costs →[2nd] lithium-ion costs are widely expected to keep falling over the next several years while small-scale TES costs are fixed-cost-dominated with no comparable learning-curve effect, so the gap identified above is more likely to widen than close in the near term →[3rd] over a much longer horizon, or at much larger scale, or with a free or low-cost heat source replacing purchased grid electricity, the capex-per-kWh penalty quantified in C2 and the round-trip-efficiency penalty quantified in C4 both shrink or become irrelevant, while lithium-ion's finite cycle life becomes the comparatively more binding constraint, and this extension contradicts no Ground Truth, it bounds the verdict to the short-duration/small-scale/grid-charged regime rather than reversing it →[3rd] the duration, scale and charging-source conditions under which each technology wins are therefore structural features of the two technologies, not a temporary cost gap expected to close with more deployment experience at the 5 MWh scale specifically

Confidence: LOW — capped by both C4 and C5 (each LOW) per the chain-head ceiling rule; this is the headline verdict chain and it inherits every unresolved cause named on C1, C3, C4 and C5. The directional finding (Li-ion dominates at this scale/duration/charging-source combination) is robust across nearly the entire bracket even though the point estimates are not — only the extreme favorable-TES/unfavorable-Li-ion tails overlap.

Conclusion C7: The 5 MWh standalone grid-charged configuration has no real-world market benchmark

GT-5 (utility TES cost model) + GT-6 (TES component structure, CSP plant context) → every molten-salt TES cost source located in this session, including GT-5 and GT-6, describes either utility or gigawatt-hour-scale plants paired with concentrating-solar collection, or industrial process-heat systems, and never a standalone megawatt-hour-class system charged from the grid [Assumes: A-19] → the absence of any such product or project across the literature searched is itself evidence that the comparison this analysis was asked to build has no real market benchmark to check against → the figures in C1 through C6 are therefore a from-scratch synthetic construction and not a benchmarked product price, and should be read as illustrating why the industry does not build systems this way rather than as a quote a developer could act on

Confidence: MEDIUM — Inputs and Rivals are clean. Inference is short: the first hop rests on [Assumes: A-19], an argument from absence that cannot be fully verified by search (a counter-example vendor or

project could exist and simply not have been found). Removing this cause would require either finding such a counter-example or a comprehensive market survey beyond this session's scope.

Conclusion C8: Thermal-only (non-electrified) delivery is a materially different, more favorable comparison

C1 (LOW, storage-only capex $1.40M central) → removing the power block and its efficiency penalty from C1′s capex, and crediting only C1′s storage-only installed cost plus the charging-heater cost used in C4 [Assumes: A-12], gives a Scenario-B capex of about $1.50M → delivering heat rather than electricity at 300 cycles/yr [Assumes: A-16] with a 90–97% thermal-retention factor [Assumes: A-2] yields about 1,395 MWh-t of delivered heat per year → annualizing that capex at 7% over 30 years [Assumes: A-11] with 2% O&M [Assumes: A-15] and dividing by delivered heat gives a levelized cost of about $108/MWh-t of delivered heat → delivered as process heat rather than re-electrified, the same 5 MWh molten-salt hardware is in a plausible range to compete against electric-resistance or fossil process heat, which is a fundamentally different comparison from the lithium-ion-battery verdict in C6 and should not be conflated with it

Confidence: LOW — capped by C1 (LOW) per the chain-head ceiling rule; offered as context, not as input to the headline Li-ion comparison, which is why it is not cited on C6′s head.

Unverified input rule (D-07), applied once for the whole section: every chain above that cites a `GT-N?` input or cites another chain rated below HIGH carries a MEDIUM or LOW confidence line naming that input/chain and, where a verification path exists, what it is; no chain in this section is rated HIGH while consuming a `GT-N?` input; every chain is rated no higher than the lowest-rated chain its head cites (C2, C4, C6 and C8 are all capped this way by C1, C3, C4/C5 and C1 respectively).

5. Abandoned Reasoning

Dead End: Using the USGS "solar salt" commodity price as the TES media price

What was tried: An early search surfaced USGS Mineral Commodity Summary figures for "solar salt" at roughly $128–153/metric ton (2021–2025) and this was briefly considered as the salt-media unit cost.

Why abandoned: "Solar salt" in the USGS mineral-commodity sense is evaporated sodium-chloride (sea/rock salt used for water softening, food, and de-icing) — a completely different material from the NaNO3/KNO3 nitrate eutectic ("solar salt" in the CSP/TES sense) this analysis is costing. The two share a name and nothing else; using the sodium-chloride price would have understated the TES media cost by roughly 5–10×.

What it ruled out: Treating any commodity-salt price index as a proxy for nitrate-eutectic TES media pricing without checking which "salt" the source actually means. GT-4 (the Reuters Events nitrate-salt range) and the NREL/SAM report's own caution about the ammonia-PPI surrogate (read in the same source as GT-5/GT-6) were used instead.

Dead End: Using the global $125/kWh Li-ion figure as the US comparison baseline

What was tried: GT-9′s $125/kWh (Ember, October 2025) was initially considered as the headline lithium-ion installed-cost figure for chain C5.

Why abandoned: The source explicitly scopes this figure to "markets outside China and the U.S."; US installed costs are reliably higher due to tariffs, labor rates, and interconnection costs. Using the single cheapest global data point as the US baseline would have manufactured an unrealistically wide competitiveness gap in lithium-ion's favor.

What it ruled out: Cherry-picking the most favorable available benchmark for one side of a comparison. GT-9 is retained in the Ground Truths list as a documented low-end global anchor, but chain C5 uses the wider, US-relevant GT-8? range instead.

Dead End: Applying the formal weighted Trade-off procedure to the TES-vs-Li-ion choice

What was tried: Considered opening the Trade-off procedure (locked criteria weights, anchored 1–5 scoring) to formalize the TES-vs-Li-ion comparison.

Why abandoned: The comparison already resolves on quantitative levelized-cost derivation chains across duration, scale, and efficiency regimes (chains C1–C8); it is not a qualitative multi-criteria option choice that benefits from locked weights and anchored scores. Forcing a weighted-score matrix onto an already-quantified cost comparison would add notation without adding information.

What it ruled out: A second, parallel scoring exercise that would have restated chain C6′s conclusion in a different, less transparent format. (Recorded again in the Techniques-not-applied process-output block.)

Dead End: Applying formal Pre-Mortem instead of Inversion as the Phase 5 adversarial technique

What was tried: The user's request explicitly asked for "inversion/pre-mortem thinking." Both techniques were considered for the Phase 5 adversarial pass.

Why abandoned: This analysis's conclusion is a claim (a cost-competitiveness verdict), not a specific deployment plan or recommendation to build a named system; the methodology's decision rule routes a claim to Inversion and a plan to Pre-Mortem. Pre-Mortem was therefore not applicable as the *formal* Phase 5 procedure, though its "assume it already failed" framing is carried in spirit through the inversion worksheet in Section 2 and the adversarial pass below.

What it ruled out: Running a plan-stress-test procedure against an artifact (a plan to build the system) that this analysis never actually produced — the analysis concludes the system is a poor choice, it does not recommend building one.

Dead End: Treating the NREL/SAM $27–35.5/kWh-t utility figure as the final answer

What was tried: Early in the research, the NREL/SAM reference-plant figure (GT-5) was the single most authoritative number found and could have been reported directly as "the" cost of molten-salt TES.

Why abandoned: The user explicitly requested a bottom-up build from constituent unit costs rather than a quoted benchmark, and GT-5 applies at a ~500×-larger scale than the 5 MWh system in question (chain C2) — reporting it unmodified would have silently answered a different question than the one asked.

What it ruled out: Substituting a correct-but-mis-scaled benchmark for the requested small-scale build. GT-5 is retained as the Phase 5 recompute/cross-check anchor in chain C2, not as the headline answer.

6. Conclusion

Recommended approach: Use the bottom-up build as the basis for decision-making, not a quoted benchmark: installed cost for the 5 MWh storage-only system comes to roughly $281/kWh_t central (range $146–454/kWh_t) (chain C1). Costed as a grid-charged electric battery, this levelizes to roughly $758/MWh delivered central (range $308–1,840/MWh) (chain C4), against roughly $162/MWh delivered central (range $78–302/MWh) for a comparable 5 MWh lithium-ion system (chain C5) — so for a short-duration, grid-electricity-charged, megawatt-hour-scale deployment, lithium-ion is the cost-competitive choice, not molten-salt TES (chain C6).

Key insight: The uncompetitiveness is not a materials or chemistry problem — the salt itself is cheap, roughly $48,000 of the ~$1.40M installed cost (chain C1) — it is two compounding scale penalties: a fixed-cost-dominated tank/balance-of-plant floor that does not shrink proportionally with capacity (chains C1, C2), and a small-turbine thermodynamic penalty that leaves a tiny power block capturing only about a third of the Carnot ceiling versus two-thirds at utility scale (chain C3). Together these are why the same technology that costs ~$27–36/kWh-t at utility scale costs roughly 9× more per kWh-t at 5 MWh (chain C2), and roughly 4.7× more per MWh delivered than lithium-ion at this scale (chain C6) — a naive comparison of raw media cost alone would have missed both penalties entirely.

Trade-offs acknowledged: This conclusion accepts LOW confidence on the exact dollar figures — the dominant cost drivers (tank fabrication, balance-of-plant, small power-block cost) are this analysis's own unverified synthesis rather than vendor quotes (chains C1, C4) — in exchange for a transparent, re-derivable bottom-up build instead of a single quoted benchmark, per the user's explicit request. It also deprioritizes

the more favorable thermal-only/process-heat use case (chain C8), which answers a different question than “vs. lithium-ion battery storage,” and sets aside the finding that no standalone grid-charged product of this type appears to exist in the market (chain C7): the verdict is necessarily a synthetic estimate illustrating why the industry does not build systems this way, not a market-benchmarked quote.

Confidence: LOW — matches the weakest contributing chains (C1, C2, C4, C5, C6 and C8, all LOW; C3 and C7 are MEDIUM and do not raise the floor). The dominant downgrade causes are GT-3? (unverified salt density, chain C1), GT-8? and GT-10? (unverified lithium-ion benchmarks, chain C5), and the unpriced [Assumes: A-5], [Assumes: A-8] and [Assumes: A-14] tank/BOP/power-block cost assumptions (chains C1, C4) — each named with its removal path (a primary thermophysical-properties read, a successful NREL-ATB read, and EPC/vendor quotes, respectively) on the chain that carries it. Despite the LOW point-estimate confidence, the directional verdict in chain C6 — lithium-ion dominates at this scale, duration and charging-source combination — is robust across nearly the entire plausible bracket; only the extreme favorable-TES/unfavorable-Li-ion tails overlap.